PROGRAMMING FUNDAMENTALS 1
SAMPLE IN CLASS EXAM
TIME : 1 HR
WEEK 6

There are two sections on this paper. **All questions are mandatory**.

Section A: (Multiple choice) is worth **30 Marks** (3 marks per question)

Section B: (Writing code) is worth **70 mark**s (4 questions)

Student Name : ____

Student Number : ______________________________________

Programme (CS/Forensics/Physics) ________________________

Group (W1/W2/W3/W4/Physics) ___________________________

Section A –Multiple choice - Please circle your answer. Only one answer applies in each question.

Section A - multiple choice. (30 Marks)

Each question is worth **3 Marks. There is exactly one correct answer in each question.**

QA1. What is true about an array of objects (e.g. String[] or Product[]) when it is first created?

A. All elements are initialised to null
B. All elements contain default values like 0 or false
C. All objects are automatically created
D. All elements contain empty objects

QA2. What happens to ArrayList indexing when an element is removed?

A. Elements after it shift left
B. Indices remain unchanged
C. The list becomes empty
D. A runtime error occurs

QA3. Which of the following is a limitation of arrays?

A. Arrays track how many elements are used
B. Arrays cannot store objects
C. Arrays automatically grow when full
D. Arrays have a fixed size once created

QA4. Utility methods are usually declared as:

A. public instance methods
B. private static final
C. public static
D. protected abstract

QA5. What is object serialization?

A. Converting objects into a stream of bytes for storage
B. Displaying objects on screen
C. Sorting objects in an ArrayList
D. Validating user input

QA6. What happens to program data stored only in memory when the program closes?

A. The data is lost because memory is volatile
B. The data is saved automatically
C. The data is converted to XML
D. The data is cached permanently

QA7. Which declaration correctly defines a constant class variable?

A. private static final int GRAVITY = 3;
B. private int gravity = 3;
C. public final int GRAVITY = 3;
D. static int gravity;

QA8. How should a static method typically be called?

A. objectName.methodName();
B. this.methodName();
C. ClassName.methodName();
D. new ClassName().methodName();

QA9. Which Javadoc tag documents a return value?

A. @version
B. @author
C. @return
D. @param

QA10. Why are utility methods often declared static?

A. They need to store object state
B. They must override superclass methods
C. They can be called without creating an object
D. They cannot return values

Section B – Long Questions (70 marks)

Answer ALL questions. Questions 1 and 2 carry 15 marks. Question 3 carries 21 Marks and Question 4 carries 19 Marks.

You should write your answers on this sheet.

Question B.1 – ArrayLists (15 Marks)

QB.1.1 Write Java code to:

1. Declare and create an ArrayList<String> called students. **(2 Marks)**

2. Add the following values:
   - "Jane Doe"
   - "John Smith"
   - "Al Jones"

(3 Marks)

QB1.2 Given an integer n, write code to

- Check that n is a valid index for ‘students’
- If so, delete the String at that position

(5 marks)

QB1.3 Using the code on previous page (to start with), write a boolean method 'findStudent(int n)'

that tries to find the String at position and returns

- true if the String was found
- false otherwise

(5 Marks)

Question B2 - Looping (15 Marks)

QB2.1 Loops with ArrayLists

Write code to print out all the values of students (as added in question **B1**) (one line per module) using :

1. While loop **(3 marks)**	2. For-each loop. **(3 Marks)**

QB2.2 Given a (populated) primitive array of integers, write a method
‘***int findSmallest(int[] a)*** ‘
Which takes in an array a and returns the position of the smallest element in the array. You can assume that the array is non-empty.

(4 Marks)

QB2.3 Using the same algorithm, write a method to find the product with the lowest unit cost using the Shop running example used in class. The method,
‘***Product findLeastExpensive()***’
Should return the product with the lowest unit Cost. You may assume that the method

a. Is written in the Store class
b. Has available to it (as a field)
‘***products***’
which is an ArrayList of Product.

You can also assume that you have the method
‘***getUnitCost***()’
available to you in the Product class (returns the unit cost of a Product object)

(5 Marks)

Question B3 - Misc (21 Marks)

QB3.1 Given the following:

```
public class Product {
    private String productName;
    private double price;
    private int quantity;
    // Calculated field
    private double totalValue;

private void calculateTotal() {
        this.totalValue = this.price * this.quantity;
    }


public void setQuantity(int quantity) {
        this.quantity = quantity;
        calculateTotal();
    }
```

This Product class has four fields. The value of `totalValue` depends entirely on the current values of ‘price and ‘quantity (`totalValue = price * quantity;`).

Describe what problems can arise from this data design. Also show how you would fix this problem. (note we need to be able to see/use what the `totalValue` is from time to time).

(5 Marks)

Given the following definition of MINUTES_PER_HOUR

```
public static final int MINUTES_PER_HOUR = 60;
```

Describe what each of the words mean in this context :

public :

static :

final :

(5 Marks)

QB3.2 Write a method that takes in three integer values (low, high, value) and returns

- **true** if the value of value is between low and high (inclusive)
- **false** otherwise

Be careful to name the method appropriately and to use appropriate variable names. **(5 Marks)**

QB3.3 Given the following code:

```
int x[] = {5, 6, 7};int i = x[0];for (int z = 1; z < x.length; z++) if (i >
x[z]) i = x[z];System.out.println(i);
```

a: What (exactly) is printed when this code is executed?

b: Rewrite the code using indentation and better variable names.

(6 Marks)

Question 4 - XML (19 Marks)

QB4.1 XML Structure (3 marks)

Consider the XML:

```xml
<student>
    <name>Alice</name>
    <course>Computing</course>
</student>
```

1. What is the root element?
2. Name two child elements.
3. State one reason why XML is useful.

(4 Marks)

QB4.2 **Try–Catch** Write Java code that:

- Asks the user to enter two integers
- Divides the first by the second
- Uses a try–catch block to handle division by zero

(6 marks)

QB4.3

Explain, using pseudocode, the steps a program would take to:

- Read an XML file
- Display all student names stored in it **(6 Marks)**

QB4.4 Match each term with its meaning, using either lines or writing it down: **(4 Marks)**

1. Create	a) Remove data
2. Read	b) Add new data
3. Update	c) Display data
4. Delete	d) Modify existing data